Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

NAME:Kavya Gada

SAP ID:60017230102

ROLL NO.:A018

BRANCH:AIML

BATCH:A1

DATE OF SUBMISSION:28/03/24

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Pg-1 60017230102

Exp-5 Kavya Gada

Aim/Objective: To implement class with members and method (static, non static, recursive and overloaded methods)

Description of Experiment:-

- Explain classes in java: In java, a class is a blueprint or template that defines the attributes and behaviour that object of that class will have. It serves as a foundation for creating objects, which are instances of the class.

- Explain objects in java: In java, an object is an instance of a class, you are essentially defining a blueprint for objects. An object is instantiated from the class, has its own set of fields and can perform actions (methods) as defined in the class.

- Syntax for declaring a class and creating an object

Declaring a class

```
public class ClassName
{
    // Fields (instance variables)
    // constructors
    // methods
}
```

- public → Access modifier
- class → Keyword indicating declaration of class
- ClassName → Name of the class

Sundaram
FOR EDUCATIONAL USE

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

2

Creating objects of the class:-

ClassName objectName = new ClassName (arguments);

- Object Name :- Name of object being created
- new : Keyword to allocate memory to object
- argument : To be passed to the constructor

- Types of variables :-

1. Local variables :-
- declared within a method
- must be intialized before use

2. Instance variables :-
- declared within a Class but outside any method, constructor or block
- associated with objects
- intialized when an object is created
- exist as long as object exists

3. ~~Class's~~ static variable :-
- declared with 'static' keyword with a class but outside any method, constructor.
- Can have different access modifiers
- belong to class rather than an instance
- Only 1 copy of class variable.

FOR EDUCATIONAL USE

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

3

- Types of functions (methods)

1. Static method:-
- associated with a class rather than any instance of that class
- Can be called using class name directly.
- Cannot access instance variables directly.

2. Non-Static / Instance method:-
- associated with instance of a class
- They can access and modify instance variables.
- Non-static methods are invoked on objects of that class

3. Recursive method:-
- call themselves directly / indirectly
- must have exit condition to prevent infinite recursion.

4. Overloaded methods:-
- methods with same name but different parameters list
- allow multiple methods having same name in a class provided parameters list is different.
- provide flexibility and code readability.

- How to access static and non-static functions:-

Static methods:- They can be directly called by class-name. No need to creat an object.

P.T.O

FOR EDUCATIONAL USE
Sundaram

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

- Types of functions (methods)

1. Static method:-
- associated with a class rather than any instance of that class
- Can be called using class name directly.
- Cannot access instance variables directly.

2. Non-Static / Instance method:-
- associated with instance of a class
- They can access and modify instance variables.
- Non-static methods are invoked on objects of that class.

3. Recursive method:-
- call themselves directly / indirectly
- must have exit condition to prevent infinite recursion.

4. Overloaded methods:-
- methods with same name but different parameters list
- allow multiple methods having same name in a class provided parameters list is different.
- provide flexibility and code readability

- How to access static and non-static functions:-

Static methods:- They can be directly called by class-name. No need to creat an object.

P.T.O

Sundaram

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Syntax :-

```
ClassName.StaticMethod();  // Accessing static Method
```

Non-static methods :-
An instance of the class must be created in order to access the non-static method.

Syntax :-
```
ClassName objectName = new ClassName();
ObjectName.nonStaticMethod();
```

- Static polymorphism (method overloading) :-
Method overloading allows a class to have multiple methods having same name but different parameter list.

eg :-
```
public class MathOperations
{ public int add (int a, int b)
  { return (a+b); }
  public int add (int a, int b, int c)
  {
    return (a+b+c); }
}

public class Main
{ public static void main (String args[])
  { MathOperations m = new MathOperation();
    int n = m.add (2, 3);
    System.out.println ("Sum : " +n); } }
```

P.T.O

FOR EDUCATIONAL USE

FOR EDUCATIONAL USE

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Output :-

Sum -: 5
Sum : 6

Conclusion: implementation of class, object, various variables and methods was understood

WAP to display area of square and rectangle with the concept of overloaded function.

```
→ import java.util.*;
Class Area
{
  public static void area (int s)
  { System.out.println ("Area : " + (s*s)); }

  public static void area (int l, int b)
  { System.out.println ("Area : " + (l*b)); }

}

Class A5002 {
public static void main (String Args[])
{ Scanner sc = new Scanner (System.in);
  System.out.println ("Enter the side of a square");
  int s=sc.nextInt();
  Area.area (s);
  System.out.println ("Enter the length and breadth
```

FOR EDUCATIONAL USE

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

6

```
int l=sc.next Int();
int b=sc.next Int();
Area = area(l,b);
}
}
```

Out put:-

```
Enter the side of a Square:
10
Area:100
Enter length and breadth
5
30
Area:150
```

Kanal
27/8/24

FOR EDUCATIONAL USE

Sundaram

FOR EDUCATIONAL USE

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Experiment No:5

WAP to perform mathematical operations on 2 complex numbers by passing and returning object as argument

Q1] Create a class employee with data member's empid, empname, designation and salary. Write methods getemployee() to take user input, showgrade() to display grade of employee based on salary, showemployee() to display details of employee.

Program :-

```
import java.util.Scanner;

public class Employee {

private String empid;

private String empname;

private String designation;

private double salary;

        public void getEmployee() {

                Scanner sc = new Scanner(System.in);

                System.out.println("Enter Employee ID: ");
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

```
        empid = sc.nextLine();
        System.out.println("Enter Employee Name: ");
        empname = sc.nextLine();
        System.out.println("Enter Employee Designation: ");
        designation = sc.nextLine();
        System.out.println("Enter Employee Salary: ");
        salary = sc.nextDouble(); sc.close();
    }
    public void showGrade() {
        char grade;
            if (salary >= 100000) {
                grade = 'A';
            }
            else if (salary >= 50000) {
                grade = 'B';
            }
            else { grade = 'C';
            }
            System.out.println("Grade: " + grade);
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

```
    }
    public void showEmployee() {

        System.out.println("Employee Details:");
        System.out.println("Employee ID: " + empid);
        System.out.println("Employee Name: " + empname);
        System.out.println("Designation: " + designation);
        System.out.println("Salary: " + salary);
    }
    public static void main(String[] args) {
        Employee employee = new Employee();
        employee.getEmployee();
        employee.showGrade();
        employee.showEmployee();
    }
}
```

Output :-

NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

```
C:\kavya\java\College Pratice Q\Exp 5 Code>javac Employee.java

C:\kavya\java\College Pratice Q\Exp 5 Code>java Employee
Enter Employee ID:
2345
Enter Employee Name:
kavya
Enter Employee Designation:
Software Devoplper
Enter Employee Salary:
200000
Grade: A
Employee Details:
Employee ID: 2345
Employee Name: kavya
Designation: Software Devoplper
Salary: 200000.0
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

WAP to display area of square and rectangle using the concept of overloaded functions

Program :-

```
 class over {
        public static void main(String[] args) {
                over obj= new over();
                obj.Area(2);
                obj.Area(2,4);
                obj.Area(2.2,3.14);
        }
        void Area(int a, int b) {
                int area = a*b;
                System.out.println("Area of rectangle = "+area);
        }
                void Area(int a) {
                 int area = a*a;
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

```
        System.out.println("Area of square = "+area);
    }
    void Area(double r, double p) {
        p=3.14;
        double area = p*r*r;
        System.out.println("Area of circle = "+area);
    }
}
```

Output :-

```
C:\kavya\java\College Pratice Q\Exp 5 Code>javac over.java

C:\kavya\java\College Pratice Q\Exp 5 Code>java over
Area of square = 4
Area of rectangle = 8
Area of circle = 15.197600000000003
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

WAP to find value of y using recursive function (static), where y=x^n

Program :-

```
import java.util.*;
class power {
        public static int power(int x, int n) {
                if(n==0) {
                        return 1;
                }
                else if(n>0) {
                        return x*power(x,n-1);
                }
                else { return 1/ power(x,-n);
        }
}
public static void main(String[] args) {
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

```
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the base: ");
        int x=sc.nextInt();
        System.out.println("Enter the power: ");
         int n=sc.nextInt();
         System.out.println(power(x,n));
        }
}
```

Output :-

```
C:\kavya\java\College Pratice Q\Exp 5 Code>java power
Enter the base:
2
Enter the power:
3
8

C:\kavya\java\College Pratice Q\Exp 5 Code>
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

WAP to count the number of objects made of a particular class using static variable and static method and display the same.

Program :-

```
public class object_counter {
static int count = 0;
        public object_counter() {
                count++;
        }
        public static int getCount() {
                return count;
         }
        public static void main(String[] args) {
                object_counter obj1 = new object_counter();
                object_counter obj2 = new object_counter();
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018

```
        object_counter obj3 = new object_counter();

        int numberOfObjects = object_counter.getCount();
        System.out.println("Number of objects created: " +
        numberOfObjects);

    }
}
```

Output :-

```
C:\kavya\java\College Pratice Q\Exp 5 Code>javac CountObject.java

C:\kavya\java\College Pratice Q\Exp 5 Code>java CountObject
Total Number of Objects: 5
C:\kavya\java\College Pratice Q\Exp 5 Code>
```


NAME:Kavya Gada

SAP ID: 60017230102

ROLL NO.:A018